

# Supplementary Material: CL-TAV Contracts, Wrappers and Decision Tables

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## Abstract

This supplement is part of the delivered manuscript package. It typesets ALG-CLTAV-02/03/04 wrappers, ALG-CLTAV-05/06/07 cores, typed records and the A/P stop tables. It does not change ALG-CLTAV-01 and does not report experiments. Supplement algorithm numbers are S1, S2, ...; they are not the main-text Algorithm 1.

## 1 Scope and numbering

Main-text Algorithm 1 is ALG-CLTAV-01. Supplement Algorithms S1–S6 are ALG-CLTAV-02, 03, 04, 05, 06 and 07. S9 in the main text calls COMMITCOMPATIBLEUPDATE (S6 / ALG-CLTAV-07), not a bare IF-HIST-UPDATE invocation.

## 2 S0–S10 duty map

| Step  | Call                           | Duty / exception                                     |
|-------|--------------------------------|------------------------------------------------------|
| S0    | INITIALIZE                     | $\Gamma, \eta, H, U$ ; distinct $k$ , charges, retry |
| S1    | IF-RESOURCE-STOP with IF-EQUIV | already decided Admit A2–A5 or P-stop                |
| S2    | PREDICTCURRENT                 | ALG-CLTAV-02; TEST-scoped GAP                        |
| S3    | SELECTANDADMIT                 | ALG-CLTAV-05; freeze $\xi$ including $U$             |
| S4    | CHARGEONCE                     | one charge; retry cap $\neq$ budget                  |
| S5    | IF-EXECUTE-RECORD              | raw record and effect class                          |
| S6    | INTERPRETOUTCOME               | ALG-CLTAV-03 + timing ALG-CLTAV-06                   |
| S7–S8 | RESOLVEOUTCOME                 | ALG-CLTAV-04; adopt returned contexts                |
| S9    | COMMITCOMPATIBLEUPDATE         | version/intersection; SPEC-ERROR; summary commit     |
| S10   | IF-RESOURCE-STOP with IF-EQUIV | P-stop or continue to S1                             |

## 3 Typed records

| Record               | Fields                                                                                   |
|----------------------|------------------------------------------------------------------------------------------|
| PredictionResult     | status OK GAP; reason; classesByTest (OK); historyVersionUsed; uncertaintyRef            |
| Decision             | kind ACTION EXIT SPEC-ERROR; actionKind; actionId; classesUsed; reason                   |
| SelectSnapshot $\xi$ | actionId; actionKind; qUsedAtSelect; historyVersion; classesUsedAtSelect; uncertaintyRef |
| ExecutionResult      | attemptId; record; effect; correlationId                                                 |
| Outcome $z$          | validity; effect; $I_z$ or NONE; summaryConfirmed; postSummary; timing                   |
| Resolution $\delta$  | control COMMIT RETRY STOP; $\Gamma'$ ; $\eta'$ ; outcome; reason                         |

## 4 Admit and post-update stops

A1 selects an affordable distinguishing TEST. A2 is KNOWN with a distinguishing TEST or eligible Prep, but no such action affordable (Stop-Budget). A3 is KNOWN with no distinguishing TEST and no eligible Prep (Stop-NoDistinguisher). A4 is UNKNOWN with no eligible Recover (Stop-Error). A5 is UNKNOWN with Recover eligible but unaffordable (Stop-Budget). A TEST prediction GAP is handled independently: a KNOWN GAP without eligible Prep is SPEC-ERROR, not A3; retry-cap remains an independent Stop-Error condition. P1 Stop-Empty; P2 Stop-Singleton; P3 Stop-645; P4 Stop-Equivalent; P5 Stop-Budget after update. Stop-NoDistinguisher is exclusive with those P-stops when no remaining distinguisher exists and no P-stop already fired. Order is Empty, Singleton, 645, Equivalent, Budget. Singleton  $h_{\text{normal}}$  is not protocol PASS.

## 5 Short arguments

Monotonicity, retention (R1)–(R5), one-step minimax and resource termination are stated in the main text. The algebraic details are the accepted method §3.9.1 arguments: update is intersection; ERROR does not exclude; minimax is a least element of a finite selectable set; billed executions are bounded by the selected mode. They are not global optimality and not independent mathematical approval.

## 6 AI disclosure fields

System: PENDING-USER (AUTH-AI). Scope (text/figures/code): PENDING-USER. Author verification: required; not completed by this sentence. IEEE AECS guidance is rechecked at submission.

## 7 Wrappers and core mechanisms

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**Algorithm S1:** PREDICTCURRENT: current-observation prediction wrapper (ALG-CLTAV-02, S2)

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**Require:** SessionContext  $\Gamma$ ; HistoryHandle  $\eta$ ; current  $H$ ; uncertainty reference  $U$ ; action library  $L$

**Ensure:** PredictionResult: status OK with nonempty classesByTest, or status GAP with a reason; the session's own  $\eta$  is read, not rebuilt

```
1  $raw \leftarrow \text{IF-PRED-OBS}(\Gamma, \eta, H, U, L);$ 
   // branch on the backend tag BEFORE reading or projecting classes; a GAP result
   // carries no classesByTest
2 if  $raw.status = \text{GAP}$  then
3   | return PredictionResult(status = GAP, reason =  $raw.reason$ );
   | // the gap stays TEST-scoped
4  $classes \leftarrow \text{PROJECTONTOCURRENTH}(raw.classes, H);$ 
   // a declared class set that projects to no nonempty class is the same named
   // gap, not OK and not  $score = 0$ 
5 if  $classes$  has no nonempty member then
6   | return PredictionResult(status = GAP, reason = no currently valid class);
7 return PredictionResult(status = OK, classesByTest =  $classes$ , historyVersionUsed =
    $\eta.version$ , uncertaintyRef =  $U$ );
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**Algorithm S2:** INTERPRETOUTCOME: execution and observation coordination (ALG-CLTAV-03, S6)

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**Input:** ExecutionResult  $er$ ; SelectSnapshot  $\xi$ ;  $\Gamma, \eta$ ; uncertainty reference  $U$

**Output:** total Outcome  $z$ , including effect, validity,  $I_z$ , and confirmation status

```

1  $z \leftarrow \text{NEWOUTCOME}()$ ; copy  $er.\text{effect}$ ,  $\xi.\text{actionKind}$  and  $\xi.\text{actionId}$  to  $z$ ;
2 if  $er.\text{effect} \in \{\text{CONFIRMED-NOT-SENT}, \text{UNKNOWN-EFFECT}\}$  then
3   return  $z$ ;
4  $(I_z, e, s, p, o, m) \leftarrow \text{IF-OBS-INTERPRET}(\Gamma, er.\text{record}, \text{clocks}, \varepsilon)$ ;
5 if  $o$  is invalid then
6   set  $z.\text{validity} \leftarrow \text{ERROR}$  and  $z.\text{effect} \leftarrow \text{UNKNOWN-EFFECT}$ ; return  $z$ ;
7  $\text{timing} \leftarrow \text{INTERPRETTIMEDOBSERVATION}(o, m, \xi, N_r, D_r, \text{Match}_r)$ ;
8 if  $\text{timing.verdict} = \text{ERROR}$  then
9   set  $z.\text{validity} \leftarrow \text{ERROR}$  and  $z.\text{effect} \leftarrow \text{UNKNOWN-EFFECT}$ ; return  $z$ ;
10 set  $z.\text{validity} \leftarrow \text{timing.verdict}$  and  $z.I_z \leftarrow I_z$ ;
11 if  $\xi.\text{actionKind} \in \{\text{PREP}, \text{RECOVER}\}$  then
12   evaluate  $\text{IF-PREP-RECOVER}(\Gamma, \eta, \xi.\text{actionId}, er.\text{record})$  into target, summary, error, target
      declaration, evidence and  $p$ ; ; // returns values only
13 set  $z.\text{summaryConfirmed} \leftarrow s$  and  $z.\text{postSummary} \leftarrow p$ ;
14 if  $e = \text{UNKNOWN-EFFECT}$ , prep error, an unconfirmed Prep/Recover result, or  $e = \text{NONE}$  with no
      confirmed summary then
15   set  $z.\text{effect} \leftarrow \text{UNKNOWN-EFFECT}$ ;
16 else
17   set  $z.\text{effect} \leftarrow e$ ;
18 return  $z$ ;
; // all  $\Gamma/\eta$  writes stay in ALG-CLTAV-04 and S9

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**Algorithm S3:** RESOLVEOUTCOME: ERROR, retry and confirmed recovery (ALG-CLTAV-04, S7–S8)

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**Require:** Outcome  $z$  (effect, kind, validity, confirmation); SelectSnapshot  $\xi$ ;  $\Gamma$ ;  $\eta$

**Ensure:** Resolution  $\delta = (\text{control}, \Gamma', \eta', \text{outcome}, \text{reason})$ :  $\Gamma'$  initialized from  $\Gamma$

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```

1  $\Gamma' \leftarrow \Gamma$ ;  $\eta' \leftarrow \eta$ ;
  // copy/persist semantics: the returned contexts are total, never undefined
2 if  $z.\text{validity} = \text{ERROR}$  or  $z.\text{effect} \in \{\text{CONFIRMED-NOT-SENT}, \text{UNKNOWN-EFFECT}\}$  or  $z.\text{prepErr}$  or
   ( $z.\text{kind}$  is Prep and  $z.\text{prepResultEvaluated}$  and  $z.\text{summaryConfirmed}$  is false) or ( $z.\text{kind}$  is
   Recover and  $z.\text{prepResultEvaluated}$  and  $z.\text{targetConfirmed}$  is false) then
  // S7 classify by kind and evaluated confirmation; do not fold
  CONFIRMED-NOT-SENT into UNKNOWN
3  do not exclude  $H$ ;
4  if  $z.\text{validity} = \text{ERROR}$  or  $z.\text{prepErr}$  or  $z.\text{effect} = \text{UNKNOWN-EFFECT}$  or ( $z.\text{kind}$  is Prep and
    $z.\text{prepResultEvaluated}$  and  $z.\text{summaryConfirmed}$  is false) then
5    mark  $\eta'$  conservative-unknown;  $\Gamma'.\text{qStatus} \leftarrow \text{UNKNOWN}$ ; clear  $\Gamma'.\text{currentSummary}$ ;
6  else if  $z.\text{effect} = \text{CONFIRMED-NOT-SENT}$  then
7    leave  $\Gamma'.\text{qStatus}$  and compatible history unchanged;
8  else
9     $\Gamma'.\text{qStatus}$  stays UNKNOWN; clear  $\Gamma'.\text{currentSummary}$ ; mark  $\eta'$  conservative-unknown;
    // unconfirmed Recover target
10  $\Gamma'.\text{retryCount} \leftarrow \Gamma.\text{retryCount} + 1$ ;
11 if  $\Gamma'.\text{retryCount} \geq \Gamma'.\text{retryCap}$  then return
   Resolution( $\text{control} = \text{STOP}, \Gamma', \eta', z, \text{reason} = \text{Stop-Error}$ );
12 return Resolution( $\text{control} = \text{RETRY}, \Gamma', \eta', z$ );
   // an ERROR/unknown outcome reaches retry, never commit
13 else if  $z.\text{kind}$  is Recover then
  // S8 a confirmed Recover is a necessary condition for this branch; sole
  writer of KNOWN; the normalized summary returns to S9
14  $\Gamma'.\text{qStatus} \leftarrow \text{KNOWN}$ ;  $z.\text{postSummary} \leftarrow (z.\text{declaredTarget}, z.\text{evidence})$ ;
15 return Resolution( $\text{control} = \text{COMMIT}, \Gamma', \eta', z$ );
   // a normal TEST/Prep with a valid observation commits; the normalized Outcome
   carries the confirmed postSummary to S9

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**Algorithm S4:** SELECTANDADMIT: test selection and resource admission (ALG-CLTAV-05, S3)

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**Require:**  $\Gamma$ ;  $\eta$ ; current  $H$ ; action library  $L$ ; typed  $pred$  (PredictionResult)

**Ensure:** a Decision: kind ACTION / EXIT / SPEC-ERROR with actionKind, actionId, classesUsed and reason

// state eligibility is independent of affordability; under UNKNOWN only an eligible Recover qualifies, never a Prep

1  $\text{eligible}(t) \leftarrow (\Gamma.\text{qStatus} = \text{UNKNOWN} ? (t.\text{kind} = \text{RECOVER} \wedge t.\text{eligibleUnknown}) : (t.\text{kind} \neq \text{RECOVER} \wedge t.\text{enabledAt} = \Gamma.\text{currentSummary}))$ ;

2  $A \leftarrow \{t \in L : \text{eligible}(t)\}$ ;

// affordability is decided by IF-SELECT-ADMIT, not assumed from eligibility

// a global or per-TEST projection gap withholds TEST scoring and keeps the gap identity; it does not itself block eligible Prep/Recover

3  $\text{testGap} \leftarrow \text{false}$ ;

4 **if**  $\text{pred.status} = \text{GAP}$  **then**

5 |  $C \leftarrow \{\}$ ;  $\text{testGap} \leftarrow \text{true}$ ;

6 **else**

7 |  $C \leftarrow \{C_t\}$  with  $C_t \leftarrow \{H \cap C : C \in \text{pred.classesByTest}[t], H \cap C \neq \emptyset\}$  for each TEST  $t \in A$ ;

8 | **if** a TEST  $t \in A$  has declared classes but  $C_t = \emptyset$  **then**

9 | |  $C \leftarrow \{\}$ ;

10 | |  $\text{testGap} \leftarrow \text{true}$ ;

// IF-SELECT-ADMIT is the single admission authority: affordability and A1-A5; exclusive kind; one-step minimax  $s(t) = \max_{C \in C_t} |C|$  over currently valid nonempty classes of affordable distinguishing TESTs  $D \cap F$  only; cost/id tie-break; never  $\text{argmin } \emptyset$ ; Prep/Recover are not observation minimax; uninformative TESTs are not A2

11  $(k, S, t^*, \text{admit}) \leftarrow \text{IF-SELECT-ADMIT}(A, \Gamma.\text{currentSummary}, C, \Gamma, \eta, H)$ ;

12 **if**  $k = \text{ADMIT}$  **then**

13 | **if**  $\text{testGap}$  and  $\text{admit} = \text{A3}$  **then return**

Decision(kind = SPEC-ERROR, reason =  $\text{pred.reason}$  or per-test prediction gap);

14 | **return** Decision(kind = EXIT, reason = the A2-A5 admission reason);

15 **else if**  $k = \text{PREP}$  **then**

16 | **return** Decision(kind = ACTION, actionKind = PREP, actionId =  $t^*$ , classesUsed =  $S$ );

17 **else if**  $k = \text{RECOVER}$  **then**

18 | **return** Decision(kind = ACTION, actionKind = RECOVER, actionId =  $t^*$ ,.);

19 **else if**  $\text{testGap}$  **then**

20 | **return** Decision(kind = SPEC-ERROR, reason =  $\text{pred.reason}$ );

21 **return** Decision(kind = ACTION, actionKind = TEST, actionId =  $t^*$ , classesUsed =  $C_{t^*}$ );

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**Algorithm S5:** INTERPRETTIMEDOBSERVATION: timed observation kernel (ALG-CLTAV-06, S6 kernel)

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**Require:** the single ownership result and the raw measurement interval from  
 IF-OBS-INTERPRET; SelectSnapshot  $\xi$ ; named obligation inputs  $N_r$  (allowed  
 interval) and  $D_r$  (physical domain); declared match policy  $\text{Match}_r$ ; declared  
 uncertainty  $\varepsilon$

**Ensure:** a TimingResult: verdict in {PASS, FAIL, INCONCLUSIVE, ERROR}; the single  
 pairing evidence; the interval used; validity

// resolve  $N_r$ ,  $D_r$  and  $\varepsilon$  from the named obligation and instance inputs, not from  
 free symbols

```

1 pair  $\leftarrow \text{Match}_r(\text{ownership}, \xi)$ ;
  // one pairing result feeds T2 and the no-response rule
2 if pair is ambiguous then return TimingResult(verdict = ERROR);
3 if pair is a legitimate trigger/response pair then
4    $M \leftarrow \text{measurementInterval} \cap D_r$ ;
5   if  $M = \emptyset$  then return TimingResult(verdict = ERROR);
6   if  $M \subseteq N_r$  then
7     return TimingResult(verdict = PASS, evidence = pair,  $M$ );
8   else if  $M \cap N_r = \emptyset$  then
9     return TimingResult(verdict = FAIL, evidence = pair,  $M$ );
10  return TimingResult(verdict = INCONCLUSIVE, evidence = pair,  $M$ );
  // open/closed endpoints preserved
  // a missing response forms a no-response only after a still-active instance
  // reaches its deadline; this round not seeing it is not a timeout
11 if the obligation is still active and expired then
12   // use the earliest admissible elapsed horizon (a scalar), with the
13   // requirement's boundary convention
14   if  $N_r$  is closed at  $U_r$  then
15     expired  $\leftarrow \text{lowerHorizon} > U_r$ ;
16   else
17     expired  $\leftarrow \text{lowerHorizon} \geq U_r$ ;
18   if expired then return TimingResult(verdict = FAIL);
19   return TimingResult(verdict = INCONCLUSIVE);
20 return TimingResult(verdict = ERROR);
  // a missing pair is not a point observation

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**Algorithm S6:** COMMITCOMPATIBLEUPDATE: compatible update and summary commit (S6 / ALG-CLTAV-07)

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**Input:** snapshot  $\xi$ ; Outcome  $z$ ; contexts  $\Gamma, \eta, H$

**Output:** UpdateResult ( $status, \eta', H', \Gamma'$ )

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1  $\Gamma' \leftarrow \Gamma$ ;
2 if  $\xi.historyVersion \neq \eta.version$  then return SPEC-ERROR with unchanged  $\eta, H, \Gamma'$ ;
3 if  $z.I_z$  is a valid compatible class then
4    $(\eta_c, H_c) \leftarrow$  IF-HIST-UPDATE with:;
5    $\eta, H, \xi.actionId, \xi.qUsedAtSelect, \xi.classesUsedAtSelect,$ ;
6    $\xi.historyVersion, z.I_z, z.postSummary$ ;
7   if  $H_c \neq H \cap z.I_z$  then return SPEC-ERROR (intersection mismatch);
8    $\eta' \leftarrow \eta_c; H' \leftarrow H_c$ ;
9  $(\eta', H') \leftarrow$  IF-HIST-UPDATE with the same snapshot fields, NONE, and  $z.postSummary$ ;
10  $H' \leftarrow H$ ;
11 if  $\Gamma'.qStatus$  is KNOWN and  $z.summaryConfirmed$  then
12    $\Gamma'.currentSummary \leftarrow z.postSummary$ ;
12 return OK with  $(\eta', H', \Gamma')$ ;
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